Nam H. Le, Ph.D.

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RESEARCH PROFILE

Computer scientist working at the intersection of **computational intelligence**, **artificial life**, and **language-driven control of living and embodied systems**. My research asks how adaptive, intelligent behaviour emerges in evolving collectives — neural agents, swarms, and cellular ensembles — and how large language and vision–language models can be used to *steer* such systems without hand-engineered reward functions.

Three lines of work:

- **Biomorphic Intelligence.** Evolutionary transitions in individuality, collective reproduction, and the emergence of multicellular-like organisation in artificial life systems (VitaNova; *Northeast J. of Complex Systems*, 2025).
- **Embodied Generative AI.** Prompt-to-Intervention (P2I) and ZapGPT: free-form natural-language control of simulated and engineered living constructs, with semantic reward supplied by a vision–language model — the first demonstration of steering emergent cellular dynamics from language alone. Now being extended from simulation to **Xenobots**, living self-assembling constructs. Funded by the U.S. Army Research Office; patent application filed.
- **Computational Intelligence.** Neuroevolution, the Baldwin effect, social versus individual learning, and evolutionary optimisation in dynamic and rugged landscapes (GECCO, PPSN, IEEE CEC, EvoStar, ALIFE).

A fourth, applied strand — **dynamical systems and computational intelligence for financial markets** — is pursued through **BunnyQuant**, a venture I founded and built single-handedly: three production platforms serving the U.S. and Vietnamese markets, each with a live web application and a native iOS app, and each grounded in my own theoretical work on regime change in complex adaptive systems.

EDUCATION

Ph.D. in Evolutionary Computation **2016 – 2023**
University College Dublin, Ireland

Thesis: *Learning and Evolution in Dynamic Environments: Computational Metaphors for Adaptive Algorithms.*
Supervisors: Prof. Michael O'Neill and Prof. Anthony Brabazon
Natural Computing Research & Applications Group (NCRA).

M.Sc. in Computer Science **2013 – 2015**
Le Quy Don Technical University *Hanoi, Vietnam*

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow **2024 – Present**
University of Vermont, Department of Computer Science *Burlington, VT, USA*

Morphology, Evolution & Cognition Laboratory. Co-mentored by Prof. Josh Bongard (UVM) and Prof. Michael Levin (Tufts University).
Lead researcher and developer on the U.S. Army Research Office–funded project *"Giving Cells a Voice."* Designed the **Prompt-to-Intervention (P2I)** architecture — SBERT sentence embeddings feeding a multi-head 1-D convolutional policy network optimised by CMA-ES, with reward supplied by a vision–language model (Qwen2.5-72B) — enabling linguistic control of cellular dynamics without engineered rewards. Demonstrated in the **ZapGPT** system. [Code] [Video]

Current line of work (2026). Extending P2I beyond simulation to **Xenobots** — self-assembling biological constructs — eliciting prescribed locomotor behaviour from free-form natural-language instruction alone. Validated against archival behavioural recordings; live biological assays in progress with collaborators at Tufts University. A **patent application has been filed** through the University of Vermont. Manuscripts under review and in preparation. *Method and results restricted pending patent prosecution and peer review.*

Senior Research Fellow **2023 – 2024**
University of Southampton, Department of Computer Science *Southampton, UK*

Agents, Interaction and Complexity group. PIs: Prof. Richard A. Watson and Prof. Michael Levin.
Led research on **evolutionary transitions in individuality (ETIs)**. Designed and built **VitaNova**, a simulation in which evolving neural agents self-organise into flocks exhibiting **collective reproduction**, mirroring transitions to multicellularity and eusociality, with emergent role specialisation. [Paper] [Code] [Video]

Doctoral Researcher **2016 – 2022**
University College Dublin, Natural Computing Research & Applications Group *Dublin, Ireland*

Investigated the interaction between **learning and evolution** in dynamic environments. Developed **evolving self-taught neural networks** integrating genetic evolution with individual and social learning for sparse-reward settings. Showed that self-learning improves robustness under rapid environmental change, while social learning accelerates adaptation in more stable regimes.

PUBLICATIONS

Author order is alphabetical by contribution, not seniority. GECCO, PPSN, IEEE CEC, EvoStar and ALIFE are the flagship competitively reviewed venues of evolutionary computation and artificial life; full papers there are archival and rigorously peer reviewed.

Journal Articles

Nam H. Le, M. Levin, R. A. Watson, J. Bongard, C. L. Buckley.
Emergent Collective Reproduction via Evolving Neuronal Flocks.
Northeast Journal of Complex Systems, Vol. 7, No. 2, Article 8, 2025.
[Full Text] [Code]

Book Chapters

Nam H. Le, Josh Bongard.
Decoupling Representation and Learning in Genetic Programming: The LaSER Approach.
In *Genetic Programming Theory and Practice XXII*, Springer Nature Singapore, 2026.
[Preprint: arXiv:2505.17309]

Preprints

Nam H. Le.
Who Pays for the Knowledge Models Copy? Model Collapse as Rogers' Paradox.
Zenodo, 2026.
[Preprint] [Code]

Nam H. Le.
Oddworld: Relational Reasoning Emerges under Implicit Selection in Evolving Attention Agents.
Zenodo, 2026.
[Preprint]

Nam H. Le.
Organismal Agency and Rapid Adaptation: The Phenopoiesis Algorithm for Phenotype-First Evolution.
arXiv:2602.00978, February 2026.
[Preprint]

Peer-Reviewed Conference Papers — Bio-Inspired AI and Language-Based Control

K. Kannan Srinivasan, **Nam H. Le**, Josh Bongard, Douglas Blackiston.
Evolving Agents to Generate and Falsify Hypotheses of Biological Self-Assembly.
In *ALIFE 2025: Proceedings of the Artificial Life Conference*, MIT Press, Kyoto, Japan, October 2025.
[Paper]

Nam H. Le, Patrick Erickson, Yanbo Zhang, Michael Levin, Josh Bongard.
ZapGPT: Free-form Language Prompting for Simulated Cellular Control.
In *Proceedings of the 2025 8th International Conference on Computational Intelligence and Intelligent Systems (CIIS 2025)*, ACM, pp. 119–126, November 2025.
[Code] [Video]

Peer-Reviewed Conference Papers — Neuroevolution and Multi-Agent Learning

Nam H. Le, A. Brabazon, M. O'Neill.
Social Learning vs Self-teaching in a Multi-agent Neural Network System.
In *Applications of Evolutionary Computation*, Springer, 2020, pp. 354–368.

Nam H. Le. *Evolving Self-taught Neural Networks: The Baldwin Effect and the Emergence of Intelligence.*
In *AISB Symposium on AI & Games*, 2019.

Peer-Reviewed Conference Papers — Artificial Life and Evolution

Nam H. Le. *Organic Selection and Social Heredity: The Original Baldwin Effect Revisited.*
In *ALife 2019 Conference Proceedings*, MIT Press.

Peer-Reviewed Conference Papers — Learning Strategies in Evolutionary Optimisation

Nam H. Le, A. Brabazon, M. O'Neill.
Evolutionary Consequences of Learning Strategies in a Dynamic Rugged Landscape.
In *GECCO Proceedings*, ACM, 2019.

Nam H. Le, A. Brabazon, M. O'Neill.
How Learning Strategies Can Promote an Evolving Population in Dynamic Environments.
In *IEEE Congress on Evolutionary Computation (CEC)*, 2019.

Nam H. Le, A. Brabazon, M. O'Neill.
The Evolution of Self-taught Neural Networks in a Multi-agent Environment.
In *EvoStar*, 2019. *(Best Paper nominee.)*

Nam H. Le, A. Brabazon, M. O'Neill.
How the Baldwin Effect Can Guide Evolution in Dynamic Environments.
In *Theory and Practice of Natural Computing (TPNC)*, 2018.

Nam H. Le, A. Brabazon, M. O'Neill.
Adaptive Advantage of Learning Strategies: A Study Through Dynamic Landscape.
In *Parallel Problem Solving from Nature (PPSN XV)*, 2018.

Nam H. Le, A. Brabazon, M. O'Neill.
The Baldwin Effect Reconsidered Through the Prism of Social Learning.
In *IEEE Congress on Evolutionary Computation (CEC)*, 2018. *(Best Student Paper nominee.)*

Peer-Reviewed Conference Papers — Genetic Programming and Representation Learning

Nam H. Le, M. O'Neill, A. Brabazon.
Social Grammatical Evolution with Imitation Learning for Real-Valued Function Estimation.
In *IEEE Congress on Evolutionary Computation (CEC)*, 2017.

Nam H. Le, H. N. Xuan, A. Brabazon, T. P. Thi.
Complexity Measures in Genetic Programming Learning: A Brief Review.
In *IEEE Congress on Evolutionary Computation (CEC)*, 2016.

Workshop, Companion and Short Papers

Nam H. Le, Patrick Erickson, Yanbo Zhang, Michael Levin, Josh Bongard.
Giving Simulated Cells a Voice: Evolving Prompt-to-Intervention Models for Cellular Control.
In *GECCO Companion Proceedings* (poster track), ACM, 2025.
[PDF] [Video]

Nam H. Le. *Evolution and Self-teaching in Neural Networks.*
In *GECCO Companion Proceedings* (poster track), ACM, 2019.

TECHNOLOGY TRANSFER AND INTELLECTUAL PROPERTY

Patent Application Filed **2026**
University of Vermont / Tufts University

Language-directed control of biological constructs, arising from Invention Disclosure D1047. Co-inventors: Prof. Josh Bongard (University of Vermont), Dr. Douglas Blackiston (Tufts University). Patent counsel retained jointly by UVM and Tufts; inter-institutional agreement in place. *Title and application number withheld pending prosecution.*

Open-source research software. All principal research systems are released publicly to support reproducibility: ALife_ZapGPT, Evolving-Neuronal-Flocks, Evolving-Self-taught-Architectures.

TEACHING EXPERIENCE

Guest Lecturer **2023 – 2024**
University of Southampton *Southampton, UK*

Delivered lectures on *Machine Learning Algorithms in Computational Finance* and *Evolutionary Algorithms.*

Graduate Instructor (M.Sc. level) **2018 – 2019**
UCD Michael Smurfit Graduate Business School *Dublin, Ireland*

Taught machine learning and data-analysis programming to master's students.

Graduate Teaching Assistant **2014 – 2015**
Le Quy Don Technical University *Hanoi, Vietnam*

Assisted in teaching *Computer Network Design* and *Network Security* to undergraduates.

Lecturer **2013 – 2014**
Hanoi University of Industry *Hanoi, Vietnam*

Taught *Fundamentals of Informatics* and *Information Systems for Accounting* to undergraduates.

Teaching philosophy. I teach adaptive systems by having students *build* them: every course I design centres on a running implementation project, so that theory is validated against behaviour the student can observe.

Courses I am prepared to teach.

- Core: Introduction to AI; Machine Learning; Data Structures & Algorithms; Python for Data Science.
- Specialist: *Computational Intelligence: Evolutionary and Nature-Inspired Algorithms*; *Optimisation*; *Multi-Agent Systems.*
- New courses I would propose: *Artificial Life and Biomorphic Intelligence*; *Foundation Models for Embodied and Biological Control*; *Machine Learning for Financial Markets.*

RESEARCH SUPERVISION AND MENTORSHIP

Krishna Kannan Srinivasan (University of Vermont). Day-to-day research mentor, outside any formal supervisory appointment. I contributed the evolving-flock framework the project was built on and directed the research design; the work became a full paper at *ALIFE 2025* (MIT Press, Kyoto), on which I am a principal co-author.

Additionally supervised junior researchers and student projects on machine learning and artificial life during doctoral and postdoctoral appointments at UCD, Southampton and UVM.

GRANTS, FELLOWSHIPS, HONOURS AND AWARDS

Outstanding Student Award **2019**
Society for Evolutionary Computation in Europe *Leipzig, Germany*

Best Paper Nomination **2019**
EvoStar Conference *Leipzig, Germany*
"*The Evolution of Self-taught Neural Networks in a Multi-agent Environment.*"

Best Student Paper Nomination **2018**
IEEE CEC / WCCI *Rio de Janeiro, Brazil*
"*The Baldwin Effect Reconsidered Through the Prism of Social Learning.*"

In preparation. Branco Weiss Fellowship (deadline January 2027).

INVITED TALKS, ACADEMIC VISITS AND INTERNATIONAL ENGAGEMENT

Invited Speaker **May 2024**
Allen Discovery Center, Tufts University (host: Prof. Michael Levin) *Medford, MA, USA*
"*Evolutionary Transitions in Individuality and the VitaNova Platform.*" [Recording]

Visiting Researcher **May 2024**
BEACON Center for the Study of Evolution in Action, Michigan State University *East Lansing, MI, USA*

Visiting Researcher **May 2024**
Simon Levin Lab, Dept. of Ecology and Evolutionary Biology, Princeton University *Princeton, NJ, USA*
Presented the VitaNova work on collective reproduction and ETIs to Prof. Simon Levin's group.

International research network. Prof. Josh Bongard (UVM) and Prof. Michael Levin (Tufts) — current co-mentors and co-authors; Dr. Douglas Blackiston (Tufts) — co-inventor, biological experimentation; Prof. Richard A. Watson (Southampton) — former PI and co-author;
Prof. Una-May O'Reilly (MIT CSAIL); Prof. Xin Yao.

PROFESSIONAL SERVICE

Peer review. *Cognitive Science* (Wiley / Cognitive Science Society); *Artificial Life* (MIT Press); *IEEE Transactions on Evolutionary Computation*; Genetic and Evolutionary Computation Conference (GECCO); IEEE Congress on Evolutionary Computation (CEC); Artificial Life Conference (ALIFE).

Science communication. Long-form essays on science, technology and society for a Vietnamese readership of approximately 20,000.

ENTREPRENEURIAL EXPERIENCE AND TECHNOLOGY TRANSLATION

Founder and Developer
BunnyQuant

Founder and sole developer of a quantitative market-analytics venture comprising **three production platforms**. **Each is deployed as a live web application with a companion native iOS app**, served by a shared research codebase:

- **bunnyquant.com** — **United States**: equities, cryptocurrencies, futures, and precious metals (gold/silver) for intraday and short-horizon trading.
- **bunnyquant.vn** — **Vietnam**: listed equities (spot) and **index derivatives/futures**, with an additional fundamental-analysis engine (DCF, Monte Carlo valuation, financial-statement parsing, peer-relative valuation) built for the local market.
- **theta.bunnyquant.com** — **Theta: Market Phase Transitions**, the applied outcome of my own dynamical-systems research (below), covering Vietnamese equities and derivatives, cryptocurrencies, and U.S. equities and futures.

Research contribution. Theta implements an original dynamical-systems treatment of market regime change developed for this venture: it identifies the regime an instrument occupies and whether that regime is building or breaking, resolves multiple timeframes jointly as one estimation problem, and yields a finite, exhaustive taxonomy of transition events — including the

transition that begins and fails, which conventional methods discard as noise. Validated across equity indices, index futures and cryptocurrencies. *Method proprietary and unpublished; details available under NDA.*

Engineering. Sole engineer across the full stack: quantitative research, Python/FastAPI backends, Next.js front ends, React Native and SwiftUI iOS clients, PostgreSQL with row-level security, background scanning with push alerting, subscription billing, and Linux/nginx deployment and operations. Data infrastructure built end to end — Databento CME tick data, FRED and CBOE for the U.S.; exchange feeds and financial statements for Vietnam; a single shared computational core so web and mobile clients cannot diverge numerically.

TECHNICAL SKILLS

Languages & frameworks: Python, PyTorch, TensorFlow, Keras, NumPy/Pandas, scikit-learn, Swift/SwiftUI, Git, Docker, Linux, LaTeX, MATLAB.
Methods: evolutionary algorithms (CMA-ES, GA, genetic programming, grammatical evolution), neuroevolution, reinforcement learning, sentence and vision–language embedding models, multi-agent systems, dynamical-systems modelling, time-series and information-theoretic analysis.
Infrastructure: HPC/cluster experiment management, large-scale evolutionary runs, reproducible open-source releases.

LANGUAGES

Vietnamese (native) · English (fluent)

ACADEMIC REFEREES

Josh Bongard — *Professor — Postdoctoral Advisor*
Department of Computer Science, University of Vermont
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Michael A. Levin — *Distinguished Professor — Postdoctoral Advisor*
Allen Discovery Center, Tufts University
michael.levin@tufts.edu · +1 (617) 627-6161 · https://as.tufts.edu/biology/people/faculty/michael-levin

Michael O'Neill — *Professor — Ph.D. Supervisor*
Director, Natural Computing Research & Applications Group, University College Dublin
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Richard A. Watson — *Professor — Postdoctoral Supervisor*
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